

C clients

Concurrent
Queries



...



Dispatch query to a worker thread

Per-query search flow

Start from the entry point

Beam-search
node expansion

Convergence-Aware
Early-Termination (§4.3)

Top-k not
converged

Popularity-Adaptive
Cache Manager (§4.2)

Node not
in cache

Update cache (CLOCK)

Top-k
converged

Update candidate list
by their PQ vectors in RAM

Re-rank with
full-precision vector

Return top-k results

Node already in cache

SSD layer

Shared SSD queue

SSD Data

